

QUBO Formulation: Generalized Assignment Problem

Generalized Assignment Problem

Problem Statement. Given a set of N jobs and M machines, assign every job to exactly one machine. Each assignment of job i to machine j yields a profit p_{ij} and consumes c_{ij} units of capacity from machine j , which has a total capacity limit C_j . The objective is to maximize the total profit across all assignments while ensuring that no machine exceeds its capacity.

Decision Variables. Let $x_{ij} \in \{0, 1\}$ for $i \in \{0, \dots, N-1\}$ and $j \in \{0, \dots, M-1\}$, where $x_{ij} = 1$ if job i is assigned to machine j , and $x_{ij} = 0$ otherwise.

QUBO Derivation. We convert the maximization problem into a minimization problem by negating the objective function. The original objective is $\max \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} p_{ij}x_{ij}$, which becomes $\min - \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} p_{ij}x_{ij}$.

The constraints are:

$$\sum_{j=0}^{M-1} x_{ij} = 1, \quad \forall i \in \{0, \dots, N-1\}, \quad (1)$$

$$\sum_{i=0}^{N-1} c_{ij}x_{ij} \leq C_j, \quad \forall j \in \{0, \dots, M-1\}. \quad (2)$$

We introduce a penalty weight $\lambda > 0$ and construct the QUBO Hamiltonian $H(x)$ by adding quadratic penalty terms for each constraint.

Step 1: Equality Constraint Penalty. For the job assignment constraint, we use the squared penalty $\lambda \left(\sum_{j=0}^{M-1} x_{ij} - 1 \right)^2$. Expanding using the idempotent property $x_{ij}^2 = x_{ij}$:

$$\begin{aligned} \left(\sum_{j=0}^{M-1} x_{ij} - 1 \right)^2 &= \sum_{j=0}^{M-1} x_{ij}^2 - 2 \sum_{0 \leq j < k \leq M-1} x_{ij}x_{ik} + 1 \\ &= \sum_{j=0}^{M-1} x_{ij} - 2 \sum_{0 \leq j < k \leq M-1} x_{ij}x_{ik} + 1. \end{aligned} \quad (3)$$

Summing over all jobs i , the total equality penalty contribution is:

$$P_{\text{eq}} = \lambda \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} x_{ij} - 2\lambda \sum_{i=0}^{N-1} \sum_{0 \leq j < k \leq M-1} x_{ij} x_{ik} + \lambda N. \quad (4)$$

Step 2: Inequality Constraint Penalty. For the capacity constraint, we follow the standard template where $c_{ij} = 1$ and $C_j = 1$. The constraint simplifies to $\sum_{i=0}^{N-1} x_{ij} \leq 1$. This is violated only when two or more jobs are assigned to the same machine. The quadratic penalty that enforces this is $\lambda \sum_{i < i'} x_{ij} x_{i'j}$. Thus, the total capacity penalty is:

$$P_{\text{ineq}} = \lambda \sum_{j=0}^{M-1} \sum_{0 \leq i < i' \leq N-1} x_{ij} x_{i'j}. \quad (5)$$

Step 3: Combined Hamiltonian. Combining the negated objective with the penalty terms yields:

$$H(x) = - \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} p_{ij} x_{ij} + \lambda \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} x_{ij} - 2\lambda \sum_{i=0}^{N-1} \sum_{0 \leq j < k \leq M-1} x_{ij} x_{ik} + \lambda \sum_{j=0}^{M-1} \sum_{0 \leq i < i' \leq N-1} x_{ij} x_{i'j} + \lambda N. \quad (6)$$

Step 4: Q Matrix and Offset. We flatten the two-dimensional variables into a single index $k = iM + j$ for $k \in \{0, \dots, NM - 1\}$. Reading off the coefficients from $H(x)$, the QUBO matrix entries $Q_{k,k'}$ and offset c are:

$$Q_{k,k} = -p_{ij} + \lambda, \quad \text{for } k = iM + j, \quad (7)$$

$$Q_{k,k'} = -2\lambda, \quad \text{for } k = iM + j, k' = iM + k \text{ with } j < k, \quad (8)$$

$$Q_{k,k'} = \lambda, \quad \text{for } k = iM + j, k' = i'M + j \text{ with } i < i', \quad (9)$$

$$Q_{k,k'} = 0, \quad \text{otherwise,} \quad (10)$$

$$c = \lambda N. \quad (11)$$

Penalty Weight Justification. The penalty weight λ must be chosen large enough to dominate any potential gain from violating a constraint. The maximum possible objective improvement from assigning a job to an infeasible configuration is bounded by $\max_{i,j} |p_{ij}|$. Therefore, setting $\lambda > \max_{i,j} |p_{ij}|$ guarantees that any infeasible assignment incurs a penalty strictly greater than the maximum profit that could be gained by violating the constraint, ensuring feasibility is prioritized.

Correctness Argument. Minimizing $H(x)$ over $\{0, 1\}^{NM}$ recovers the optimal assignment because (i) all feasible solutions satisfy the constraints exactly, resulting in zero penalty energy, so their total energy equals the negated profit; (ii) any infeasible solution violates at least one constraint, incurring a penalty strictly greater than $\max_{i,j} |p_{ij}|$. Since the maximum

possible profit is bounded by $\sum_{i,j} |p_{ij}|$, and λ is chosen to exceed the per-constraint violation gain, the energy of any infeasible solution strictly exceeds that of the best feasible solution. Consequently, the global minimum of $H(x)$ must correspond to a feasible assignment that maximizes the original profit.

Illustrative Example. Consider a small instance with $N = 2$ jobs and $M = 2$ machines. The variables are $x_{00}, x_{01}, x_{10}, x_{11}$, mapped to x_0, x_1, x_2, x_3 via $k = 2i + j$. Assume profits $p_{00}, p_{01}, p_{10}, p_{11}$, and capacity parameters $c_{ij} = 1, C_j = 1$. The Hamiltonian becomes:

$$\begin{aligned} H(x) = & -p_{00}x_0 - p_{01}x_1 - p_{10}x_2 - p_{11}x_3 \\ & + \lambda(x_0 + x_1 + x_2 + x_3) \\ & - 2\lambda(x_0x_1 + x_2x_3) \\ & + \lambda(x_0x_2 + x_1x_3) + 2\lambda. \end{aligned} \tag{12}$$

The Q matrix entries are $Q_{00} = Q_{11} = Q_{22} = Q_{33} = -p_{ij} + \lambda$, $Q_{01} = Q_{23} = -2\lambda$, $Q_{02} = Q_{13} = \lambda$, and all others zero. The offset is $c = 2\lambda$. For a concrete profit instance, e.g., $p_{00} = 5, p_{01} = 3, p_{10} = 4, p_{11} = 6$ with $\lambda = 10$, the optimal feasible assignment is $x_{01} = 1, x_{10} = 1$ (jobs assigned to different machines), yielding $x = (0, 1, 1, 0)$ and energy $H(x) = 33$. Any assignment violating capacity or job constraints yields strictly higher energy.